

Freight Rate Prediction: Approach and Results

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Summary

A LightGBM gradient-boosting model predicts posted_rate from load features and calendar features. On a chronological holdout it beats a lane-median baseline by 43% on MAE (113.5 vs 199.5) and 4% on RMSE (634.4 vs 660.6) across all holdout rows. On typical loads (rate per mile inside 0.91 to 3.41, 9,458 of 9,600 holdout rows) MAE is 58.1 vs 145.7 for the baseline. A small number of extreme rates dominate the overall RMSE; see the limitations section.

Train / test split and validation approach

- Labelled data (48,000 rows, 2025-01-01 to 2025-10-31) was sorted by date. The first 37,881 rows fit the model; the most recent 9,600 rows (2025-08-31 to 2025-10-31) form the holdout.
- Why chronological: the supplied validation set is a later period (2025-11-01 to 2025-12-31), so a random split would leak future information and overstate accuracy.
- Encoders and imputers are fitted on the fit partition only, then applied to the holdout, the validation set and the December inputs.
- Boosting rounds are chosen by early stopping on the holdout (best round 455, 555 trained). The final model is refit on all labelled rows with that round count.
- A lane-median baseline (median rate per pickup-to-delivery lane, overall median for unseen lanes) is scored on the same holdout as the reference the model must beat.

Data quality findings and how they were handled

- train: 48,000 rows, 14 columns
- date range: 2025-01-01 to 2025-10-31 (unparseable dates: 0)
- duplicate rows: 0; duplicate load IDs: 0
- missing: weight=300, market_index=374
- non-positive distance: 0; non-positive weight: 292
- validation: 12,000 rows, 13 columns
- date range: 2025-11-01 to 2025-12-31 (unparseable dates: 0)
- duplicate rows: 0; duplicate load IDs: 0
- missing: weight=165, market_index=249
- non-positive distance: 0; non-positive weight: 145
- december: 31 rows, 11 columns
- date range: 2025-12-01 to 2025-12-31 (unparseable dates: 0)

- duplicate rows: 0; duplicate load IDs: n/a (no load_id column)
- missing: none
- non-positive distance: 0; non-positive weight: 0
- target posted_rate: min=57.22, median=2030.76, max=25533.00; non-positive=0
- target quantiles 1% / 50% / 99% / 99.9%: 327.2 / 2,030.8 / 5,972.8 / 12,854.6
- rate per mile quantiles 0.1% / 1% / 50% / 99% / 99.9%: 0.44 / 1.67 / 2.15 / 3.18 / 10.07
- median rate per mile by weekday (Mon..Sun): 2.13, 2.14, 2.16, 2.17, 2.16, 2.13, 2.13
- median rate per mile by month: 1:2.03, 2:2.06, 3:2.14, 4:2.15, 5:2.19, 6:2.25, 7:2.19, 8:2.12, 9:2.16, 10:2.16
- extreme rows (rate per mile outside 0.92..3.41): 663 of 48,000 (1.4%)
- extreme rows, share of each equipment type: Dry Van 1.3%, Flatbed 1.4%, Reefer 1.5%
- extreme rows, share by month: 1:1.6%, 2:1.3%, 3:1.4%, 4:1.3%, 5:1.2%, 6:1.3%, 7:1.3%, 8:1.4%, 9:1.3%, 10:1.6%
- extreme rows, median distance: typical 954.00 vs extreme 880.20
- extreme rows, median weight: typical 31,443.50 vs extreme 30,754.00
- extreme rows, median market_index: typical 1.06 vs extreme 1.05
- extreme rows, median quote_signal: typical 2.06 vs extreme 2.04
- extreme rows, rate per mile as a multiple of the typical median: median 2.2x (10th-90th pct 0.2x-4.4x)
- equipment only in validation: none
- places with more than one pickup latitude in train: 0
- places with more than one delivery latitude in train: 0
- missing coordinates before restoring -> after: train: 0 -> 0, validation: 0 -> 0, december: 124 -> 0
- Fixes applied:
 - missing coordinates are restored from the same place elsewhere in the training data; remaining numeric gaps are median-imputed and categorical gaps mode-imputed (both fitted on the training partition only); unseen categories are ignored by the encoder.
 - Duplicates are reported, not deleted, because repeated loads can be valid observations.
 - Holdout: the worst 1% of rows account for 97% of the squared error (RMSE 634.4 vs MAE 113.5).
 - Median absolute error: 40.7; worst single error: 16,322.8 (see holdout_worst_errors.csv).
 - Calendar features' share of model gain: 0.2%.
 - Top features by gain: distance 92%, pickup_lon 3%, delivery_lon 2%, equipment_Dry Van 1%, quote_signal 1%, market_index 0%, equipment_Reefer 0%, weight 0%.
 - Typical holdout rows (9,458 of 9,600, rate per mile inside 0.91..3.41): LightGBM RMSE 84.8 / MAE 58.1 vs baseline RMSE 216.0 / MAE 145.7.
 - Outlier rule: 519 training rows dropped because rate per mile fell outside 0.91..3.41 (Q1 - 3*IQR .. Q3 + 3*IQR of the fit partition); treated as noisy labels that no feature explains. Early stopping watched typical holdout rows only. Holdout and validation rows are never dropped.
 - December inputs lack these model features, so they were filled with the typical value from the last 30 days of labelled data: ['market_index', 'quote_signal'].

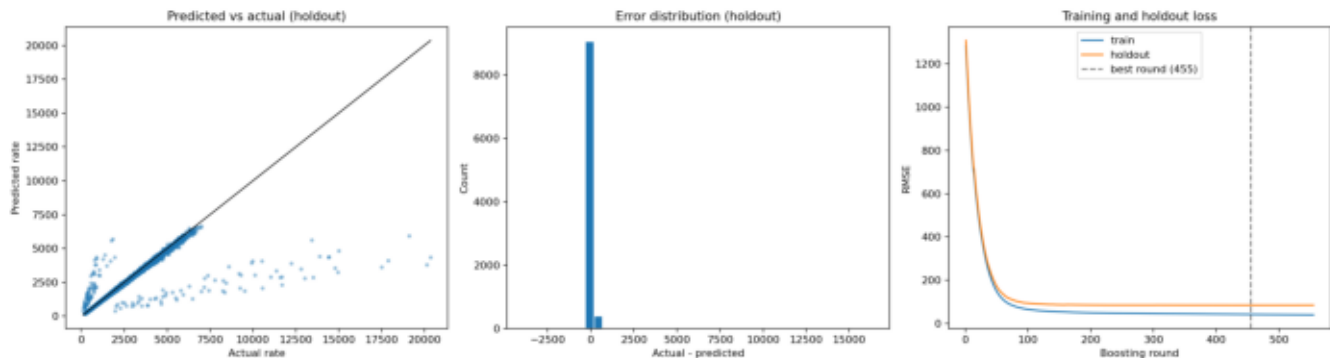
- Split: validation is a later period than training, so training rows were ordered by date; the first 80% fit the model and the last 20% form the holdout.

Model choice

LightGBM was chosen because the data is tabular, mixes numeric and categorical inputs, and rates depend non-linearly on distance, equipment and lane. Trees need no feature scaling and handle these interactions directly. Categories are one-hot encoded. Features are the supplied columns plus calendar features (day of week, day of month, month, days to and since the nearest US federal holiday). The December chart varies only by date, so calendar features are what allow the model to respond to it.

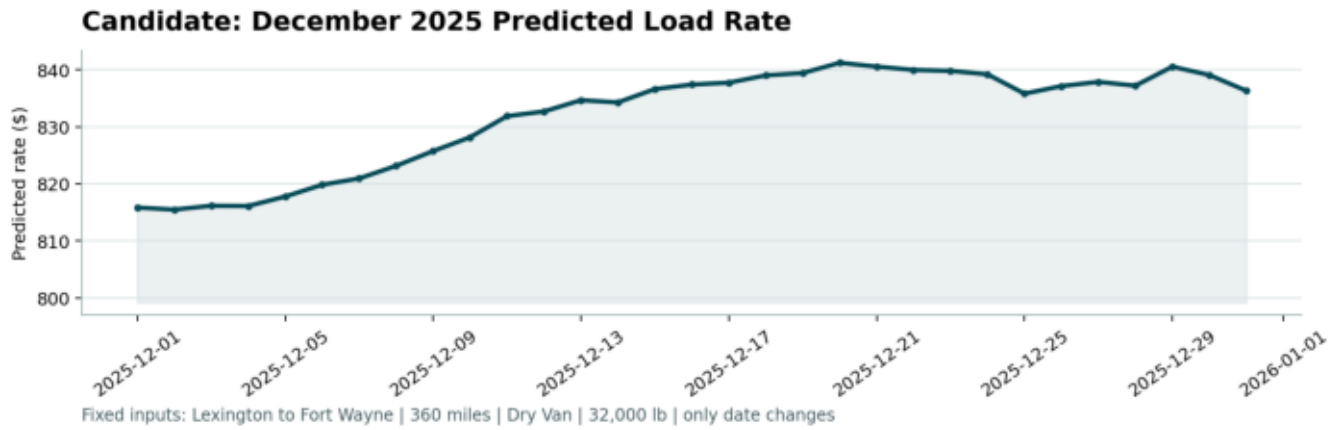
Holdout results

Model	Rows	RMSE	MAE
LightGBM	All holdout rows (9,600)	634.4	113.5
Lane-median baseline	All holdout rows (9,600)	660.6	199.5
LightGBM	Typical rows (9,458)	84.8	58.1
Lane-median baseline	Typical rows (9,458)	216.0	145.7



Holdout diagnostics: predicted vs actual, error distribution, and training/holdout loss by boosting round.

December 2025 prediction chart (produced by score.py)



Fixed inputs: Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb; only the date changes.

Predicted rates range from \$815.5 to \$841.2 across December. Missing December features (for example market signals) are filled with the typical value from the last 30 days of labelled data.

Limitations

- Training data ends 2025-10-31, while validation runs to 2025-12-31. The model has never seen November or December, so it cannot learn their seasonality directly. Holiday-distance features let it reuse patterns from earlier holidays, but that is an assumption, not something the data proves.
- About 1.5% of holdout rows have an extreme rate per mile. They produce most of the squared error, so overall RMSE stays high while typical-load error is far lower. Both are reported rather than hiding the difference.
- The holdout covers the most recent labelled months, which may not match conditions in the validation period.